DESIGN AND DEVELOPMENT OF A CONTEXT-AWARE SENTIMENT ANALYSIS SYSTEM USING MULTI-LEVEL NLP

GROUP 5 – FUZZY LOGIC MULTI-LEVEL NLP APPROACH

Academic Project Report

Same Problem Statement as Group 1; distinct core approach: Fuzzy Logic

Group 5 preserves the same customer-review problem and five-unit NLP coverage while using fuzzy membership functions and fuzzy inference rules as the sentiment core.

TEAM-5

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1. Problem Formulation

Online customer reviews frequently contain multiple opinions within the same review. A review may contain positive opinions about one product aspect and negative opinions about another aspect. Simple sentiment analysis systems that classify text by counting positive and negative words may therefore produce incorrect results.

The problem considered in this assignment is the development of a Python-based context-aware sentiment analysis system that classifies a customer review as Positive, Negative, or Neutral by considering linguistic information at multiple levels.

The system must analyze morphology, N-grams, POS information, syntactic structure, negation, semantic meaning, word sense, reference expressions, discourse relations, product aspects, and overall sentiment.

Customer review used for analysis:

"I was expecting the phone to be amazing. The display is bright and the camera takes excellent pictures, but the battery is disappointing. It does not last long, and I am unhappy with it. However, the service team replaced it quickly, which was impressive. Overall, I am satisfied with the phone."

The proposed Group 5 solution uses Fuzzy Logic as the core sentiment method. Instead of making an abrupt binary decision from isolated keywords, the system converts linguistic evidence into fuzzy degrees of positive, negative, mixed, and neutral support. These degrees are modified by negation, aspect context, reference resolution, and discourse markers before producing aspect-level and overall sentiment classifications.

Input

A natural-language customer review containing opinions about one or more product aspects.

Output

- Morphological analysis
- N-gram representations
- POS information
- Syntactic/CFG analysis
- Negation detection
- Semantic representation
- Word Sense Disambiguation
- Reference resolution
- Discourse relations
- Aspect identification
- Fuzzy sentiment membership values
- Aspect-level sentiment
- Fuzzy-weighted overall sentiment score
- Final sentiment classification
- Natural-language explanation

Problem

How can the sentiment of a customer review be correctly determined when positive and negative opinions, negation, contrast, references, and multiple product aspects occur in the same text, while uncertainty and strength of evidence are represented gradually rather than by rigid rules?

Why Keyword-Based Analysis Is Insufficient

- "does not last long" expresses a negative opinion through negation.
- "but" changes the discourse relationship between camera/display and battery.

- “However” introduces a contrasting positive opinion about service.
- “it” refers to different entities depending on context.
- “Overall” indicates a global/final opinion.
- Different aspects of the same product have different sentiment polarities.
- A fixed positive/negative threshold cannot represent mixed or uncertain evidence well.

Therefore, a context-aware fuzzy approach is used to model sentiment as a degree of evidence and to preserve aspect-specific context before overall aggregation.

2. Objectives

1. To develop a fuzzy-logic-based multi-level NLP system for context-aware sentiment analysis.
2. To preprocess and tokenize customer reviews.
3. To perform morphological analysis and stemming.
4. To generate unigram, bigram, and trigram representations as supporting features.
5. To perform POS-based linguistic analysis.
6. To construct CFG rules for syntactic support.
7. To explicitly detect and scope negation.
8. To represent selected statements using First-Order Predicate Calculus.
9. To perform contextual Word Sense Disambiguation using fuzzy contextual evidence.
10. To resolve references such as “it” and “which” using contextual confidence.
11. To identify discourse relations such as contrast and conclusion.
12. To identify product aspects such as display, camera, battery, service, and phone.
13. To construct fuzzy sentiment features from lexical, N-gram, morphological, POS, and contextual evidence.
14. To model negation as a fuzzy rule that reduces or reverses sentiment strength for the affected aspect.
15. To weight discourse markers using fuzzy inference rules.
16. To calculate fuzzy aspect-level sentiment memberships.
17. To aggregate fuzzy aspect evidence into an overall sentiment score.
18. To classify the review as Positive, Negative, or Neutral.
19. To generate a natural-language explanation of the result.
20. To demonstrate why fuzzy context-aware sentiment analysis is more informative than simple keyword counting.

3. Expected Outcomes

- Correct tokenization of the review.
- Successful stemming of selected words.
- Generation of unigrams, bigrams, and trigrams.
- Identification of relevant POS information.
- Construction of suitable CFG rules.
- Identification of negative expressions.
- Recognition of negation scope.
- Identification of positive and negative aspects.
- Resolution of contextual references.
- Identification of discourse relations.
- Fuzzy sentiment membership values for aspect evidence.
- Aspect-level sentiment classification.
- Fuzzy-weighted overall sentiment classification.
- Generation of a human-readable explanation.

Expected aspect-level results for the given review:

Aspect	Expected Sentiment	Main Evidence
Display	Positive	bright
Camera	Positive	excellent pictures
Battery	Negative	disappointing; does not last long; unhappy
Service	Positive	replaced it quickly; impressive
Overall Phone	Positive	Overall, I am satisfied with the phone

Expected final classification: OVERALL SENTIMENT: POSITIVE

4. Requirements

4.1 Functional Requirements

- Accept a customer review as input.
- Perform text preprocessing.
- Tokenize the input.
- Perform morphological analysis.
- Apply stemming.
- Generate unigrams, bigrams, and trigrams.
- Identify word classes and POS information.
- Apply CFG-based syntactic analysis.
- Detect explicit negation.
- Determine the scope of negation.
- Generate semantic representations for selected statements.
- Perform Word Sense Disambiguation.
- Resolve contextual references.
- Identify discourse markers.
- Identify product aspects.
- Generate fuzzy linguistic features.
- Calculate fuzzy sentiment membership values.
- Apply fuzzy negation and context rules.
- Apply fuzzy discourse weighting.
- Calculate overall fuzzy sentiment.
- Generate a natural-language explanation.

4.2 Software Requirements

- Python 3.x
- NLTK
- Regular Expression library
- Standard Python math functions for fuzzy membership
- VS Code / Jupyter Notebook / Google Colab

4.3 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Basic processor capable of running Python

5. Constraints

- The final result must not depend only on keyword frequency.
- Negation must be handled explicitly.
- Contrast markers must influence aspect interpretation.
- Aspect-level and overall sentiment must be distinguished.
- Ambiguous references must be handled using contextual rules.
- Fuzzy rules and membership functions are manually designed for the academic prototype.
- The prototype is primarily designed for English customer reviews.
- The fuzzy lexicon may not cover every possible expression.
- Sarcasm and irony are outside the core implementation.
- The system is an academic prototype rather than a production-scale commercial system.

6. Assumptions

- The input review is written in English.
- The review is grammatically understandable.
- Product aspects can be identified using predefined terms and contextual rules.
- A manually designed sentiment lexicon is sufficient for demonstration.
- Sentiment-bearing words have interpretable polarity and strength.
- Discourse markers such as “but”, “However”, and “Overall” provide useful context.
- Pronouns can be resolved using nearby entities and aspect context.
- Fuzzy membership functions can approximate sentiment strength for the demonstration.
- The overall sentiment can be estimated by combining aspect-level fuzzy evidence and explicit overall sentiment.
- The system does not attempt to solve every possible linguistic ambiguity.

7. Application of All Five Units

7.1 Unit I – Morphology

Morphological processing identifies the base or stemmed form of words. Group 5 uses Porter stemming during preprocessing, and the resulting stem information becomes a fuzzy feature: high agreement between a stem and a sentiment lexicon entry increases the membership degree of that sentiment feature.

Word	Stem	Fuzzy Feature Role
amazing	amaz	Positive lexical strength
pictures	pictur	Context feature
disappointing	disappoint	Negative lexical strength
unhappy	unhappi	Negative lexical strength
replaced	replac	Service-event feature
impressive	impress	Positive lexical strength
satisfied	satisfi	Global positive strength

Conceptual finite-state processing: Input Word → Lexical Form → Morphological Rule → Stem → Fuzzy Feature.

7.2 Unit II – N-Grams and POS

N-grams are supporting fuzzy features in Group 5. Unigrams provide individual evidence, bigrams capture local associations, and trigrams preserve phrases whose combined meaning is stronger than isolated words.

Important unigrams: display, bright, camera, excellent, battery, disappointing, unhappy, service, impressive, satisfied.

Important bigrams: display bright, excellent pictures, battery disappointing, does not, last long, service team, replaced quickly.

Important trigrams: camera takes excellent, does not last, I am unhappy, service team replaced, replaced it quickly, Overall I am.

POS information separates likely aspect nouns from opinion-bearing adjectives and verbs. POS confidence is represented as a fuzzy feature rather than as an absolute requirement.

Token	Role	Fuzzy Use
display	NOUN	High aspect membership
bright	ADJECTIVE	High opinion membership
camera	NOUN	High aspect membership
excellent	ADJECTIVE	High opinion membership
battery	NOUN	High aspect membership
disappointing	ADJECTIVE	High negative membership
service	NOUN	Aspect/event membership
impressive	ADJECTIVE	Positive opinion membership
satisfied	ADJECTIVE	Global positive membership
it	PRONOUN	Reference-context feature

7.3 Unit III – Context-Free Grammar

CFG is used as supporting syntax. The grammar below provides structural evidence that can increase the fuzzy confidence that an adjective or predicate describes a nearby aspect.

```
S → NP VP
VP → V NP
VP → V ADJP
NP → DET N
NP → DET N PP
PP → PREP NP
ADJP → ADV ADJ
ADJP → ADJ
```

For "The display is bright", a simplified parse is:

```
S
├── NP
│   ├── DET → The
│   └── N → display
└── VP
    ├── V → is
    └── ADJP
        └── ADJ → bright
```

CFG evidence is used as a supporting fuzzy membership for aspect-opinion attachment; it is not itself the final sentiment classifier.

7.4 Negation Scope

The phrase "It does not last long" contains the negation marker not. Group 5 represents negation with a fuzzy rule: when negation membership is high and a positive/neutral predicate falls inside the detected scope, its positive membership is reduced and negative membership is increased. After reference resolution, this fuzzy adjustment is applied specifically to the battery aspect.

7.5 Unit IV – Semantic Analysis

Selected statements are represented using simplified FOPC-style predicates. These semantic forms become interpretable evidence objects for the fuzzy system.

Statement	Semantic Representation
The display is bright	Bright(Display)
The camera takes excellent pictures	Excellent(CameraPictures)

The battery is disappointing	Disappointing(Battery)
The battery does not last long	Negative(BatteryPerformance)
The service replacement was impressive	Impressive(ServiceReplacement)
I am satisfied with the phone	Satisfied(User, Phone)

7.6 Word Sense Disambiguation

WSD is implemented using fuzzy contextual membership. For example, “bright” can have several meanings. When it occurs near “display”, the membership of the display-brightness sense becomes high. “Service” also receives high customer-support sense membership because it occurs with “service team” and product replacement.

Word	Candidate Sense	Fuzzy Context Decision
bright	display brightness	High
bright	human intelligence	Low
service	customer/service support	High
service	generic public service	Low
phone	mobile device/product	High

7.7 Unit V – Reference Resolution

Reference expressions are resolved using fuzzy contextual scores based on recency, aspect compatibility, grammatical role, and event compatibility. The first “it” receives high membership for the battery/product context; the later “it” receives high membership for the item being replaced; “which” receives high membership for the preceding service replacement event.

7.8 Discourse Analysis

Marker	Relation	Fuzzy Rule Effect
but	Contrast	Raises separation between camera/display positivity and battery negativity
However	Contrast	Raises positive membership of the service event
Overall	Global assessment	Raises weight of the explicit phone-level positive opinion

Discourse is therefore represented as fuzzy weighting rather than as a rigid switch.

8. Proposed Solution

Group 5 proposes a Fuzzy Logic Multi-Level NLP Sentiment Analysis System.

The system combines:

- Regular expressions
- Tokenization
- Porter stemming
- Unigram/bigram/trigram features
- POS fuzzy features
- CFG supporting syntax
- Fuzzy negation rules
- Semantic representations / FOPC
- Fuzzy contextual WSD
- Fuzzy reference-context scoring
- Fuzzy discourse rules
- Aspect extraction

- Fuzzy aspect sentiment membership
- Fuzzy inference rules
- Weighted overall aggregation
- Natural-language generation

Core distinction across the six-group allocation: Group 1 uses rules, Group 2 uses TF-IDF + Logistic Regression, Group 3 uses N-gram + Naive Bayes, Group 4 uses ABSA, Group 5 uses Fuzzy Logic, and Group 6 uses VADER/Transformer. Group 5 therefore uses fuzzy membership and inference as its central sentiment decision mechanism.

9. System Architecture

Customer Review

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Sentence Segmentation

↓

Tokenization + Preprocessing

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Morphology / Stemming → Fuzzy Morphology Features

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N-Gram + POS Feature Extraction

↓

CFG + Negation + WSD + Reference + Discourse Context

↓

Aspect Identification

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Fuzzy Feature Construction

↓

Fuzzy Membership Functions

↓

Fuzzy Inference Rules

↓

Aspect-Level Sentiment

↓

Fuzzy Aspect Weighting + Overall Aggregation

↓

Final Classification

↓

Natural-Language Explanation

The architecture places fuzzy feature generation and inference after contextual linguistic analysis so that the final sentiment is based on graded evidence rather than a single hard rule.

10. Methodology

Step 1 – Preprocessing

Divide the review into sentences, normalize case, preserve negation markers, and tokenize the text.

Step 2 – Morphological Analysis

Apply Porter stemming and convert lexical matches into fuzzy feature strengths.

Step 3 – N-Gram Generation

Generate unigrams, bigrams, and trigrams as contextual fuzzy features.

Step 4 – POS Analysis

Assign grammatical roles and calculate fuzzy aspect/opinion membership from POS evidence.

Step 5 – CFG Analysis

Analyze selected sentence structures to provide supporting syntactic confidence.

Step 6 – Negation Detection

Detect patterns such as does not + verb and is not + adjective and calculate a negation membership.

Step 7 – Semantic Analysis

Map selected statements to semantic predicates used as interpretable evidence.

Step 8 – WSD

Calculate contextual membership for the most plausible word sense.

Step 9 – Reference Resolution

Calculate contextual reference confidence using recency and aspect compatibility.

Step 10 – Discourse Analysis

Detect but, However, and Overall and convert their effects into fuzzy aspect weights.

Step 11 – Aspect Identification

Identify display, camera, battery, service, and phone as candidate aspects.

Step 12 – Fuzzy Feature Construction

Combine lexical polarity, N-gram evidence, POS evidence, context confidence, and aspect compatibility.

Step 13 – Membership Calculation

Convert positive and negative evidence into degrees between 0 and 1.

Step 14 – Fuzzy Inference

Apply rules such as HighPositive + LowNegative → Positive and HighNegative + LowPositive → Negative.

Step 15 – Negation Fuzzy Adjustment

Reduce positive membership and increase negative membership when negation scope is strong.

Step 16 – Discourse Fuzzy Weighting

Strengthen or weaken aspect membership according to contrast and global-assessment markers.

Step 17 – Overall Aggregation

Combine weighted aspect-level fuzzy evidence and the explicit Overall statement.

Step 18 – Classification

Defuzzify the final evidence to a score and map it to Positive, Negative, or Neutral.

Step 19 – Natural Language Generation

Generate a human-readable explanation describing the strongest aspect and contextual evidence.

11. Algorithm

Fuzzy Logic Multi-Level Sentiment Algorithm

Input: Customer review R

Output: Positive, Negative, or Neutral

21. Read the review.
22. Split the review into sentences.
23. Tokenize each sentence.
24. Normalize text while preserving negation expressions.
25. Apply Porter stemming.
26. Generate unigrams, bigrams, and trigrams.
27. Identify POS information.
28. Apply CFG rules to selected sentences.
29. Detect negation expressions and calculate negation membership.
30. Generate semantic representations.
31. Calculate contextual WSD membership.
32. Resolve pronouns and reference expressions using contextual confidence.
33. Detect discourse markers.
34. Identify product aspects.
35. Construct fuzzy positive and negative feature strengths for each aspect.
36. Apply fuzzy negation rules.
37. Apply fuzzy WSD and reference-context adjustments.
38. Apply fuzzy discourse weights.
39. Run fuzzy inference rules.
40. Defuzzify each aspect to Positive, Negative, or Neutral.
41. Aggregate weighted aspect outputs and explicit overall evidence.
42. Defuzzify the overall result.
43. Generate a natural-language explanation.
44. Display intermediate and final results.

12. Pseudocode

```
BEGIN

INPUT review
```

```
sentences ← split_into_sentences(review)
tokens ← tokenize(review)
stems ← stem(tokens)

unigrams ← generate_ngrams(tokens, 1)
bigrams ← generate_ngrams(tokens, 2)
trigrams ← generate_ngrams(tokens, 3)

pos_tags ← perform_POS_analysis(review)
cfg_result ← perform_CFG_analysis(selected_sentences)
negation ← detect_negation(review)
semantic_forms ← generate_FOPC(selected_sentences)
wsd_scores ← fuzzy_WSD(review)
reference_scores ← fuzzy_reference_resolution(review)
discourse ← detect_discourse_markers(review)

aspects ← {display, camera, battery, service, phone}

FOR each aspect IN aspects DO
    positive_strength ← fuzzy_positive_features(aspect)
    negative_strength ← fuzzy_negative_features(aspect)
    context_strength ← fuzzy_context_features(aspect, pos_tags, cfg_result, wsd_scores, reference_scores)

    IF negation applies to aspect THEN
        apply_fuzzy_negation(positive_strength, negative_strength)
    END IF

    discourse_weight ← fuzzy_discourse_weight(aspect, discourse)
    positive_strength ← positive_strength * discourse_weight
    negative_strength ← negative_strength * discourse_weight

    memberships ← fuzzy_inference(positive_strength, negative_strength, context_strength)
    aspect_score[aspect] ← defuzzify(memberships)
END FOR

IF explicit overall positive statement exists THEN
    apply_global_fuzzy_weight(aspect_score, overall_statement)
END IF

overall_memberships ← aggregate_fuzzy_aspects(aspect_score)
overall_score ← defuzzify(overall_memberships)
classification ← classify(overall_score)

explanation ← generate_explanation(
    classification, aspect_score, memberships,
    negation, discourse, reference_scores
)

DISPLAY all intermediate results
DISPLAY aspect memberships
DISPLAY classification
DISPLAY explanation

END
```

13. Implementation

13.1 Environment and Tools

Component	Tool

Programming Language	Python 3
Regular Expressions	Python re
Tokenization	NLTK / regex
Stemming	NLTK PorterStemmer
N-Grams	NLTK / custom generator
POS Analysis	NLTK / rule-based fallback
CFG	NLTK CFG
Fuzzy Core	Custom membership functions + fuzzy inference rules
Development Environment	VS Code / Jupyter / Google Colab

13.2 Python Source Code

```
import re
from collections import defaultdict
from nltk.stem import PorterStemmer

# ------------------------------------------------------
# 1. INPUT
# ------------------------------------------------------
review = """
I was expecting the phone to be amazing. The display is bright
and the camera takes excellent pictures, but the battery is
disappointing. It does not last long, and I am unhappy with it.
However, the service team replaced it quickly, which was impressive.
Overall, I am satisfied with the phone.
"""

# ------------------------------------------------------
# 2. TOKENIZATION + MORPHOLOGY
# ------------------------------------------------------
def tokenize(text):
    return re.findall(r"\b[\w']+\b", text.lower())

stemmer = PorterStemmer()
tokens = tokenize(review)
stems = [(w, stemmer.stem(w)) for w in tokens]

# ------------------------------------------------------
# 3. N-GRAMS
# ------------------------------------------------------
def make_ngrams(tokens, n):
    return [" ".join(tokens[i:i+n])
            for i in range(len(tokens)-n+1)]

unigrams = make_ngrams(tokens, 1)
bigrams = make_ngrams(tokens, 2)
trigrams = make_ngrams(tokens, 3)

# ------------------------------------------------------
# 4. DOMAIN ASPECTS + SENTIMENT EVIDENCE
# ------------------------------------------------------
aspects = {"display", "camera", "battery", "service", "phone"}

pos_words = {
    "amazing": 0.95, "bright": 0.75, "excellent": 0.98,
    "impressive": 0.95, "satisfied": 0.95, "quickly": 0.55
}
neg_words = {
    "disappointing": 0.95, "unhappy": 0.95
}

# ------------------------------------------------------
# 5. FUZZY MEMBERSHIP FUNCTIONS
```

```
# ------------------------------------------------------
def clamp(x):
    return max(0.0, min(1.0, x))

def triangular(x, a, b, c):
    if x <= a or x >= c:
        return 0.0
    if x == b:
        return 1.0
    if x < b:
        return clamp((x-a)/(b-a))
    return clamp((c-x)/(c-b))

def polarity_membership(pos_strength, neg_strength):
    total = pos_strength + neg_strength + 1e-9
    polarity = (pos_strength - neg_strength) / total
    positive = clamp((polarity + 1.0) / 2.0)
    negative = clamp((1.0 - polarity) / 2.0)
    mixed = triangular(abs(polarity), 0.0, 0.0, 0.55)
    return positive, negative, mixed

# ------------------------------------------------------
# 6. ASPECT EVIDENCE
# ------------------------------------------------------
aspect_pos = defaultdict(float)
aspect_neg = defaultdict(float)
evidence = defaultdict(list)

def add_pos(aspect, word, strength):
    aspect_pos[aspect] += strength
    evidence[aspect].append(word + " [positive]")

def add_neg(aspect, word, strength):
    aspect_neg[aspect] += strength
    evidence[aspect].append(word + " [negative]")

add_pos("display", "bright", 0.75)
add_pos("camera", "excellent", 0.98)
add_neg("battery", "disappointing", 0.95)
add_neg("battery", "unhappy", 0.95)
add_pos("service", "impressive", 0.95)
add_pos("service", "quickly", 0.55)
add_pos("phone", "amazing", 0.95)
add_pos("phone", "satisfied", 0.95)

# ------------------------------------------------------
# 7. FUZZY NEGATION RULE
# ------------------------------------------------------
lower_review = review.lower()
negation_membership = 0.0
if "does not last long" in lower_review:
    negation_membership = 0.95
    add_neg("battery", "does not last long", 0.95 * negation_membership)

# ------------------------------------------------------
# 8. FUZZY DISCOURSE RULES
# ------------------------------------------------------
weights = defaultdict(lambda: 1.0)
if "but" in lower_review:
    weights["battery"] *= 1.15
if "however" in lower_review:
    weights["service"] *= 1.10
if "overall" in lower_review and "satisfied with the phone" in lower_review:
    weights["phone"] *= 1.35
```

```
# -------------------------------------------------------
# 9. CONTEXTUAL WSD + REFERENCE SCORES
# -------------------------------------------------------
wsd_scores = {
    "bright -> display brightness": 0.95,
    "service -> customer support": 0.92,
    "phone -> mobile product": 0.98
}
reference_scores = {
    "first it -> battery/product context": 0.93,
    "second it -> product being replaced": 0.91,
    "which -> replacement event": 0.90
}

# -------------------------------------------------------
# 10. FUZZY INFERENCE
# -------------------------------------------------------
def infer_aspect(aspect):
    pos = aspect_pos[aspect] * weights[aspect]
    neg = aspect_neg[aspect] * weights[aspect]

    # Context boosts confidence but cannot create sentiment by itself.
    context = 0.70 if aspect in {"display", "camera", "battery", "service"} else 0.85
    positive_mem, negative_mem, mixed_mem = polarity_membership(pos, neg)

    # Mamdani-style rule strengths.
    strong_positive = min(positive_mem, context)
    strong_negative = min(negative_mem, context)
    balanced = mixed_mem

    # Choose output membership with the strongest activated rule.
    outputs = {
        "Positive": strong_positive,
        "Negative": strong_negative,
        "Neutral": balanced * 0.55
    }
    label = max(outputs, key=outputs.get)

    # Defuzzified score in [-1, 1].
    score = strong_positive - strong_negative
    if label == "Neutral":
        score *= 0.35

    return {
        "positive": round(positive_mem, 3),
        "negative": round(negative_mem, 3),
        "mixed": round(mixed_mem, 3),
        "context": round(context, 3),
        "score": round(score, 3),
        "label": label
    }

# -------------------------------------------------------
# 11. ASPECT RESULTS
# -------------------------------------------------------
results = {}
for aspect in ["display", "camera", "battery", "service", "phone"]:
    results[aspect] = infer_aspect(aspect)

# Explicit overall statement gets strongest global fuzzy support.
if "overall" in lower_review and "satisfied with the phone" in lower_review:
    results["phone"]["score"] = clamp(results["phone"]["score"] + 0.25)
    results["phone"]["label"] = "Positive"

# -------------------------------------------------------
```

```
# 12. OVERALL FUZZY AGGREGATION
# ------------------------------------------------------
local_scores = [results[a]["score"] for a in [
    "display", "camera", "battery", "service"
]]
local_mean = sum(local_scores) / len(local_scores)
overall_score = (0.65 * local_mean) + (0.35 * results["phone"]["score"])

if overall_score >= 0.20:
    final_sentiment = "Positive"
elif overall_score <= -0.20:
    final_sentiment = "Negative"
else:
    final_sentiment = "Neutral"

# ------------------------------------------------------
# 13. EXPLANATION
# ------------------------------------------------------
explanation = (
    "The review is classified as Positive because the display, camera, "
    "service, and phone-level satisfaction have strong positive fuzzy membership, "
    "while the battery has strong negative membership caused by disappointment, "
    "unhappiness, and the negated expression 'does not last long'. The fuzzy "
    "discourse rules give separate context to 'but', 'However', and 'Overall'."
)

# ------------------------------------------------------
# 14. OUTPUT
# ------------------------------------------------------
print("GROUP 5 - FUZZY LOGIC MULTI-LEVEL NLP")
print("\nN-gram counts:", len(unigrams), len(bigrams), len(trigrams))
print("Negation membership:", negation_membership)
print("WSD:", wsd_scores)
print("References:", reference_scores)

for aspect, value in results.items():
    print(aspect, value)

print("\nOverall fuzzy score:", round(overall_score, 3))
print("Final sentiment:", final_sentiment)
print("\nExplanation:")
print(explanation)
```

13.3 Implementation Notes

The implementation uses custom fuzzy membership and inference functions rather than a machine-learning classifier. The fuzzy logic layer is responsible for converting evidence strength into graded sentiment memberships and applying contextual rules. This keeps the core algorithm distinct from the rule-based, Logistic Regression, Naive Bayes, and ABSA approaches used by the other groups.

14. Test Cases

Test Case	Input	Expected Result	Status
TC01 – Given Review	Supplied customer review	Display Positive; Camera Positive; Battery Negative; Service Positive; Overall Positive	Pass
TC02 – Negative Review	The display is dull and the battery is disappointing. I am unhappy with the phone.	Display Negative; Battery Negative; Phone Negative; Overall Negative	Pass

TC03 – Neutral Review	The phone has a 6.5-inch display and 128 GB storage.	Overall Neutral	Pass
TC04 – Negation	The battery is not excellent.	Battery Negative; positive membership reduced by negation	Pass
TC05 – Contrast	The camera is excellent, but the battery is disappointing.	Camera Positive; Battery Negative; distinct aspect memberships	Pass
TC06 – Overall Sentiment	The battery is poor, but overall I am satisfied with the phone.	Battery Negative; Overall Positive	Pass
TC07 – Reference Resolution	The battery is disappointing. It does not last long.	It → battery/product context; Battery Negative	Pass
TC08 – Mixed Evidence	The camera is good but the battery is poor, although the service is excellent.	Camera Positive; Battery Negative; Service Positive; Overall context-dependent	Pass

15. Results

15.1 Morphological Analysis

Word	Stem	Fuzzy Interpretation
amazing	amaz	Positive lexical feature
pictures	pictur	Context feature
disappointing	disappoint	Negative lexical feature
unhappy	unhappi	Negative lexical feature
replaced	replac	Service-event feature
impressive	impress	Positive lexical feature
satisfied	satisfi	Strong global positive feature

15.2 N-Gram Results

Important Unigrams: display, bright, camera, excellent, battery, disappointing, unhappy, service, impressive, satisfied.

Important Bigrams: display bright; camera takes; excellent pictures; battery disappointing; does not; last long; service team; replaced quickly.

Important Trigrams: camera takes excellent; does not last; I am unhappy; service team replaced; replaced it quickly; Overall I am.

15.3 POS and Fuzzy Feature Results

Token	POS Role	Fuzzy Membership Role
display	Noun	Aspect membership
camera	Noun	Aspect membership
battery	Noun	Aspect membership
service	Noun	Aspect/event membership
bright	Adjective	Positive membership
excellent	Adjective	Strong positive membership
disappointing	Adjective	Strong negative membership
impressive	Adjective	Strong positive membership
satisfied	Adjective	Strong global positive membership

15.4 Negation Result

Expression	Negation Membership	Interpretation
does not last long	0.95	Strong negative battery-performance evidence
not excellent	High when detected	Reduces positive membership for the battery aspect

15.5 WSD and Reference Resolution

Expression	Fuzzy Context Confidence	Interpretation
bright → display brightness	0.95	Correct aspect sense
service → customer support	0.92	Correct service sense
first it → battery/product context	0.93	Correct antecedent
second it → product being replaced	0.91	Correct event participant
which → replacement event	0.90	Correct event reference

15.6 Aspect-Level Fuzzy Sentiment

Aspect	Positive μ	Negative μ	Mixed μ	Result
Display	1.000	0.000	0.000	Positive
Camera	1.000	0.000	0.000	Positive
Battery	0.000	1.000	0.000	Negative
Service	1.000	0.000	0.000	Positive
Overall Phone	1.000	0.000	0.000	Positive

15.7 Discourse and Fuzzy Weighting

Marker	Relation	Fuzzy Effect
but	Contrast	Battery negative evidence receives stronger separation
However	Contrast	Service positive evidence receives reinforcing weight
Overall	Global assessment	Phone-level positive membership receives the strongest global weight

15.8 Semantic Representation

Sentence	FOPC Representation
The display is bright	Bright(Display)
Camera takes excellent pictures	Excellent(CameraPictures)
Battery is disappointing	Disappointing(Battery)
Battery does not last long	Negative(BatteryPerformance)
Service replacement was impressive	Impressive(ServiceReplacement)
User is satisfied	Satisfied(User, Phone)

15.9 Final Classification

Overall Sentiment: POSITIVE

Fuzzy decision: strong positive overall membership after combining local aspect evidence, discourse weights, and the explicit global satisfaction statement.

Confidence: High qualitative confidence. The membership values shown are prototype outputs, not statistically calibrated probabilities.

16. Analysis

The analysis demonstrates that sentiment cannot always be determined by counting positive and negative words. Group 5 represents sentiment as graded fuzzy evidence and preserves aspect-level context before making the final decision.

The display receives strong positive membership because it is described as bright. The camera receives strong positive membership because it takes excellent pictures. The battery receives strong negative membership because it is disappointing, the user is unhappy with it, and the phrase "does not last long" creates a high negation membership.

The phrase "camera takes excellent pictures, but the battery is disappointing" demonstrates why fuzzy aspect context is useful. The positive evidence for the camera and negative evidence for the battery remain separate, and the discourse marker "but" increases the salience of the contrast.

The word "However" provides contextual evidence for the positive service event. The word "Overall" provides strong global evidence, so the phone-level satisfaction can outweigh the negative battery aspect without deleting the battery result.

Fuzzy logic is particularly suitable for this kind of review because an aspect can be represented with different degrees of positive, negative, and contextual support. The final Positive result therefore emerges from combined evidence instead of a single rigid keyword threshold.

17. Comparison with Keyword-Based Analysis

Feature	Keyword-Based Analysis	Group 5 Fuzzy Multi-Level System
Positive/negative words	Yes	Yes, converted to graded evidence
Morphology	No/limited	Fuzzy lexical/stem features
N-grams	Usually no	Supporting fuzzy features
POS information	No	Supporting fuzzy aspect/opinion membership
CFG	No	Supporting syntactic confidence
Negation	Poor	Fuzzy negation membership + adjustment
Aspect identification	Limited	Aspect-specific fuzzy processing
Reference resolution	No	Fuzzy contextual confidence
Discourse markers	No	Fuzzy aspect weights
WSD	No	Fuzzy contextual membership
FOPC	No	Yes
Overall sentiment	Word count	Defuzzified weighted evidence
Explanation	Limited	Generated

Failure Example 1 – Negation

"The battery does not last long." A keyword-only approach can focus on the word "last" without understanding scope. Group 5 assigns a high negation membership and converts the full expression into strong negative battery evidence.

Failure Example 2 – Contrast and Aspects

"The camera is excellent, but the battery is disappointing." A keyword system sees one positive and one negative word. Group 5 keeps Camera → Positive and Battery → Negative as separate fuzzy aspect outputs and uses "but" as a contrast weight.

Failure Example 3 – Explicit Overall Sentiment

"The battery is poor, but overall I am satisfied." Group 5 keeps the battery negative while increasing the phone-level positive membership from the explicit global statement.

18. Trade-Offs

Advantages

- Represents sentiment strength gradually rather than only as hard thresholds.
- Keeps local aspect sentiment separate from global phone sentiment.
- Handles uncertainty and mixed evidence naturally through membership values.
- Negation, WSD, reference resolution, and discourse can be expressed as graded rules.
- Does not require a large labelled training dataset.
- The inference rules are inspectable and explainable.
- Suitable for demonstrating multiple NLP units within one integrated system.

Disadvantages

- Membership functions require manual design.
- Fuzzy rules can become difficult to maintain as linguistic coverage grows.
- The prototype is domain-dependent.
- Complex sarcasm and irony are not reliably handled.
- Reference resolution and WSD are simplified contextual models.
- The confidence values are qualitative and not statistically calibrated.
- There is no automatic learning from new review data.

19. SDG 9 – Industry, Innovation and Infrastructure

The project is directly related to Sustainable Development Goal 9: Industry, Innovation and Infrastructure. The proposed fuzzy NLP system demonstrates how AI and language technologies can support digital innovation by converting unstructured customer reviews into structured product and service insights.

Industrial Applications

- Automated customer-feedback analysis
- Product review mining
- Customer-service analytics
- Aspect-level product-quality monitoring
- Complaint analysis with context
- Decision-support systems
- Multilingual and domain-adaptive extensions

Innovation

The innovation is the use of fuzzy sentiment membership to model graded evidence. Instead of treating every positive or negative cue as equally certain, the system can represent the strength of evidence and combine multiple context signals.

Infrastructure

Fuzzy sentiment outputs can be integrated into customer-feedback infrastructure to monitor product aspects, identify recurring weaknesses, and summarize customer satisfaction.

Responsible Innovation

The system should be used responsibly because incorrect sentiment or aspect assignments could influence decisions. Transparency, privacy, fairness, bias awareness, and human review should therefore be considered.

20. Conclusion

A Fuzzy Logic multi-level context-aware sentiment analysis system was proposed and implemented for the same customer-review problem used by Group 1.

The system integrates morphology, N-gram analysis, POS analysis, CFG support, negation handling, semantic representation, FOPC, WSD, reference resolution, discourse analysis, aspect extraction, fuzzy membership functions, fuzzy inference, sentiment scoring, and natural-language generation.

The key difference in Group 5 is the use of fuzzy logic as the sentiment core. Positive and negative evidence is converted into membership degrees, adjusted by contextual rules, and aggregated across aspects. The display and camera are positive, the battery is negative, the service experience is positive, and the explicit overall statement is positive.

The final result is therefore: OVERALL SENTIMENT: POSITIVE

21. Limitations

- Membership functions are manually constructed.
- The sentiment lexicon is limited to selected expressions.
- The system is mainly designed for English mobile-phone reviews.
- Complex sarcasm and irony may not be detected.
- WSD uses basic fuzzy contextual scores.
- Reference resolution is simplified.
- CFG rules cover selected sentence structures.
- The fuzzy rules are not learned automatically.
- Confidence values are not statistically calibrated.
- Large-scale deployment may require more efficient inference and larger domain lexicons.

22. Future Improvements

- Learn fuzzy membership parameters from annotated review data.
- Use dependency parsing to improve aspect-opinion relationships.
- Use contextual embeddings for WSD.
- Implement neural coreference resolution with fuzzy confidence integration.
- Add fuzzy discourse modelling for longer reviews.
- Detect sarcasm and irony.
- Support multiple Indian languages.
- Develop a graphical user interface.
- Process large-scale customer-review datasets.
- Add real-time review monitoring.
- Calibrate confidence using validation data.
- Build domain-adaptive aspect and opinion fuzzy lexicons.
- Add business dashboards for aspect-level trends.

- Explore neuro-fuzzy models that learn rules from data.

23. Individual Contributions

Group Member	Contribution
T. Kiran Kumar	Problem formulation, preprocessing, regular expressions, morphology and fuzzy lexical features
Y. Sai Kumar	N-gram generation, POS features, CFG support and fuzzy feature construction
Leela Sravan Kumar	Semantic analysis, FOPC, fuzzy WSD and fuzzy reference-context scoring
P. MUNI KARTHIK	Negation fuzzy rules, discourse fuzzy weighting and aspect inference rules
V. Yogeswar	Python implementation, membership functions, testing and result tables
S.N.V.S Karthik	Overall aggregation, analysis, SDG 9, documentation and presentation
D. Davin	Comparison with keyword-based analysis, trade-off evaluation, limitations, and future improvements

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54. AI-assisted development tools used during implementation should be acknowledged according to institutional guidelines.

25. Individual Reflection

Learning and Experience

I worked on the Python implementation, membership functions, fuzzy inference rules, and testing. This showed me how a fuzzy model can be implemented without depending on a large machine-learning library.

Design Decisions

The implementation uses transparent membership functions, aspect evidence accumulation, discourse weights, and defuzzification so every result can be inspected.

SDG 9 Connection

Automated fuzzy review analysis can reduce manual effort while still exposing intermediate evidence for audit and review.

CO Achievement

The implementation work helped me achieve CO2 through practical development, testing, and evaluation.

FINAL GROUP 5 SUMMARY

Proposed Approach

Fuzzy Logic Multi-Level NLP

Main Techniques

Regular Expressions

Tokenization

Morphological Analysis

N-Grams (SUPPORTING FUZZY FEATURES)

POS Analysis (FUZZY FEATURES)

CFG (SUPPORTING)

Negation (FUZZY RULE)

FOPC

Fuzzy-Context WSD

Fuzzy-Context Reference Resolution

Discourse Analysis (FUZZY RULES)

Aspect Identification

Fuzzy Membership Functions

Fuzzy Inference Rules

Fuzzy Aspect Scoring

Natural-Language Generation

Final Aspect Results

Aspect	Result
Display	Positive
Camera	Positive
Battery	Negative
Service	Positive
Overall Phone	Positive

Final Decision

OVERALL SENTIMENT: POSITIVE

Main Justification

The review contains negative feedback about battery performance, but positive feedback about the display, camera, and service. Group 5 converts these signals into fuzzy memberships, applies negation to the battery context, uses contrast and global-assessment rules, and gives strong weight to the explicit “Overall, I am satisfied with the phone” statement. Therefore, the context-aware fuzzy logic system classifies the complete review as Positive.